

MAST32001 Week 2 Summary

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This week in the courses, we learn two parts: how random numbers are made, how they are used.

Lecture 3 was about random number generation. It first compared physical randomness with pseudo random generates (PRNGs). Physical sources, such as coin tosses, radioactive decay, atmospheric noise or thermal noise, are truly random. But they cannot be repeated. A PRNG is a fixed machine. It has an internal state s , an update rule $S_{n+1} = TS_n$, an output function, and a seed. The seed is part of the computational rule, so we can repeat the same result. We studied linear congruential generator $X_{n+1} = (aX_n + c) \bmod m$. Their period is bounded by the modulus m . But a long period alone is not enough. A good generator must also give roughly uniform marginals, weak dependence, and good behavior for high-dimensional tuples. So we should use established generators such as R's Mersenne-Twister, not write our own LCG. Throughout, $U(0, 1)$ uniform variables are the basic building block for all other distributions.

From uniforms, we learned several ways to build other distributions. By the probability integral transform, the inverse transform method gives $X = F^{-1}(U)$ for $U \sim U(0, 1)$. Example: exponential $X = -\log(U)/\lambda$. For discrete distributions, we split $(0, 1)$ at the cumulative probabilities. We check generators by histograms over the theoretical density, Q-Q plots, the empirical CDF, and successive-pair scatter plots. Correct marginals alone do not guarantee independence. Instead of inventing a new algorithm for each distribution, we used known identities: e^X is lognormal; χ^2 is chi-square for standard normal; $\gamma(1, \lambda)$ is exponential; a ratio of independent Gammas gives a Beta; $Z/\sqrt{V/\nu}$ is Student's t; and two independent chi-squares form an F-statistic. Box-Muller uses the polar form of the bivariate normal to turn two uniforms into two independent $N(0, 1)$ draws. Acceptance-rejection handles cases where inversion is not convenient. With a proposal g satisfying $f(x) \leq cg(x)$, we draw $Y \sim g$ and accept it with probability $f(Y)/(cg(Y))$. The expected number of proposals per accepted value is c . The proposal must have tails at least as heavy as the target's. The worked example is a normal target with a Cauchy proposal.

Lecture 4 developed Monte Carlo methods as a computational principle: difficult expectations or integrals are estimated by repeated simulation. If $X_1, \dots, X_m$ are independent draws from density f , then $\theta = E[h(x)]$ is generated by the sample mean $\hat{\theta} = \frac{1}{m} \sum_{i=1}^m h(X_i)$. The strong law of large number gives consistency. $Var(\hat{\theta}) = \sigma_h^2/m$ gives the $m^{-1/2}$ convergence rate. The CLT justifies approximate intervals $\hat{\theta} \pm 1.96 \frac{s}{\sqrt{m}}$. We must separate statistical sampling error, which comes from a finite observed data set, from Monte Carlo error, which comes from a finite number of simulation draws. Increasing m reduces only the latter. Examples included integrating x^4 and $\sin(x)$ over finite intervals, extending the idea to higher dimensions, and the classic estimate of π from points inside a unit quarter-circle. To halve the Monte Carlo standard error, we need four times m , so brute-force precision is expensive.

The same machinery is used in statistical studies. By generating many data sets, we estimate bias, variance and MSE through the bias-variance decomposition, the coverage probability of confidence intervals, which is a Bernoulli proportion with its own Monte Carlo standard error, and the power of tests. We also recognize that a simulation study is itself a statistical experiment. Importance sampling was introduced as variance-reduction technique. It rewrites $I = E_g \frac{h(X)}{g(X)}$ with a proposal g placed where the integrand contributes strongly. The final topic was multi-variate simulation. Reproducing correct marginals is not enough, because dependence between coordinates must also be captured. Spherically symmetric vectors decompose into a radius $R = ||X||_2$ and a direction $U = X/R$ uniform on $\mathbb{S}^{P-1}$. Generating $Z \sim N_p(0, I_p)$ and setting $U = \frac{Z}{||Z||_2}$ gives a uniform direction, because the standard normal is rotationally invariant. Elliptical distributions arise as $X = \mu + RAU$ with shape ΣAA^t . So using a Cholesky or eigen decomposition from Lecture 2, we generate $X = \mu + AZ$ with $Z \sim N_p(0, I_p)$. The normal case has $R^2 \sim \chi_p^2$, and heavier radial laws produce heavier tailed elliptical distributions.

The inverse transform method for simple distributions such as the exponential, together with practical R calls like `runif()`, `rnorm()`, `set.seed()`, `qnorm()` and `ecdf()`, felt intuitive and straightforward. The most difficult material was putting together the assumptions behind acceptance-rejection, the envelope $f(X) = cg(X)$, the expected acceptance rate $1/c$, and the need for a heavier tailed proposal, with the rotational invariance argument that justifies drawing uniform directions on the sphere by normalizing a multivariate normal.